

The empirical recurrence for OEIS A279659, recovered from its tail

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Abstract

OEIS A279659 records “Empirical recurrence of order 51” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 2208$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 54$. Its last coefficient is $c_{51} = 134217728000 \neq 0$, so no further tail satisfies anything shorter; any order-51 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A279659 is “Number of $3 \times n$ 0..2 arrays with no element equal to a strict majority of its horizontal and antidiagonal neighbors, with the exception of exactly two elements, and with new values introduced in order 0 sequentially upwards.”. It has offset 1 and begins

0, 24, 734, 19986, 498424, 11256083, 239891281, 4888317479, ...

The entry states:

Empirical recurrence of order 51 (see link above)

As of the “Last modified” line on the live entry (Dec 16 2016 11:47:31, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 2208$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 2208$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 51: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 51.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 4416$ exact terms from $n = 3$ on, it returns order 51 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{51} c_i a(n-i),$$

c_1	45	c_{14}	−1951218522	c_{27}	−547466354096	c_{40}	8135388241920
c_2	−627	c_{15}	2202162098	c_{28}	9652999927296	c_{41}	20062725390336
c_3	1446	c_{16}	3975531090	c_{29}	26331325145184	c_{42}	23247187869696
c_4	25617	c_{17}	−30704962884	c_{30}	−21319415993120	c_{43}	14617558450176
c_5	−109959	c_{18}	61215896968	c_{31}	−47981612486592	c_{44}	−5473123565568
c_6	−151148	c_{19}	−35517676560	c_{32}	−18122386415808	c_{45}	499923550208
c_7	2089389	c_{20}	−105687303948	c_{33}	10085419754816	c_{46}	−7496332738560
c_8	−4481367	c_{21}	379919762696	c_{34}	65386732182912	c_{47}	914085642240
c_9	−10791016	c_{22}	−715795578120	c_{35}	68719816383744	c_{48}	−1351404748800
c_{10}	66160989	c_{23}	1256764953144	c_{36}	−8845102331904	c_{49}	870737510400
c_{11}	−92867301	c_{24}	186579586296	c_{37}	−39417378948096	c_{50}	0
c_{12}	−140269915	c_{25}	−4105562956800	c_{38}	−65141375686656	c_{51}	134217728000
c_{13}	913319334	c_{26}	35578947072	c_{39}	−41523465052160		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 54$, and it is the recurrence OEIS A279659 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{51} - c_1 x^{51-1} - \dots - c_{51}$. Its constant term is $-c_{51} = -134217728000$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 51 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 51, so $\deg q = 51$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 51, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 51, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 134217728000.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-51 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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